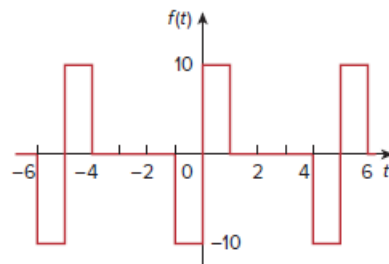


Problem

The signal displayed by a medical device can be approximated by the waveform shown in Fig. 17.86. Find the Fourier series representation of the signal.

Figure 17.86:



Step-by-step solution

Step 1 of 1

$$T = 5$$

$$\omega_0 = \frac{2\pi}{T} = \frac{2\pi}{5}$$

$$f(t) = \text{odd function} \quad a_0 = 0 = a_n$$

$$b_n = \frac{4}{T} \int_0^{T/2} f(t) \sin(n\omega_0 t) dt = \frac{4}{5} \int_0^{10} 10 \sin(0.4n\pi t) dt$$

$$= \frac{-8 \times 5}{2\pi n} \cos(0.4n\pi t) \Big|_0^1$$

$$= \frac{20}{n\pi} [1 - \cos(0.4n\pi)]$$

$$f(t) = \frac{20}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} [1 - \cos(0.4n\pi)] \sin 0.4n\pi t$$